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Assignment-1

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1) TASK:1 - Machine learning systems in a retail Bank:

a) Loan Default Prediction:

- learning paradigm: Supervised learning.
- learning Task: classification.
- T(Task): Predict whether a customer will default.
- P(Performance): Accuracy, Precision & recall.

36/50

Question paper missing. Many calculation mistake noted in each question.

b) Credit card Spending Prediction:

- learning paradigm: supervised learning.
- Learning Task: Regression
- T: Predict next month spending.

2) TASK 2: Bayis theorem - Medical Screening:

Given, $P(D) = 0.02$, $P(+|D) = 0.95$, specificity = 0.90, $P(+|\neg D) = 0.10$

To Find: $P(D|+)$

Formula: $P(D|+) = [P(+|D) \times P(D)] / P(+)$

$P(D|+) = [0.95 \times 0.02] / [(0.95) \times (0.02)] + [0.10 \times 0.98)]$

$P(D|+) = 0.95 \times 0.02 / 0.117$

$= 0.019 / 0.117$

$= 0.1624$

Result: The probability of actually having the disease after testing positive is 16.24%.

3) TASK 3: Joint Probability

Given,

X	Y=0	Y=1
0	0.10	0.20
1	0.15	0.25
2	0.20	0.10

a) Marginal Distribution:

For X:

$P(X=0) = 0.10 + 0.20 = 0.30$

$P(X=1) = 0.15 + 0.25 = 0.40$

$P(X=2) = 0.20 + 0.10 = 0.30$

For Y:

$P(Y=0) = 0.10 + 0.15 + 0.2$

$P(Y=1) = 0.20 + 0.25 + 0.$

b) Expected values & variance:

$E(X) = \Sigma x P(x)$

$= 0(0.30) + 1(0.40) + 2(0.30)$

$= 1$

$E(X^2) = 0 + 0.40 + 1.20 = 1.60$

$Var(X) = E(X^2) - [E(X)]^2$

$= 1.60 - 1^2$

$= 0.60$

$E(Y) = 0(0.45) + 1(0.55)$

$= 0.55$

4) Task 4: Normal distribution and MLE:

a) Given: $X \sim N(50, 16)$

Mean = 50

S.D = $\sqrt{16} = 4$

i) $P(X=52) \Rightarrow Z = (X-\mu)/\sigma$

$= 1 - \phi 2$

$= 2.28\%$

ii) $P(46 < X < 54) \Rightarrow Z_1 = (46-50)/4 =$

$Z_2 = (54-50)/4 =$

$\Rightarrow \phi 1 - \phi(-1)$

$= 68.26\%$

Given readings: 12, 15, 11, 14, 13

Mean = $\mu = \Sigma x/n = 65/5 = 13$

Variance = 2. / Unbaished variance = 2.5.

ASK:5 - Entropy & Information Theory:

Given: $P = \{1/2, 1/4, 1/8, 1/8\}$

Formulae: $H(X) = -\Sigma P(x) \log_2 P(x)$

Solution: $H(X) = -[(1/2)\log_2(1/2) + (1/4)\log_2(1/4) + (1/8)\log_2(1/8) + (1/8)\log_2(1/8)]$

$= 0.5 + 0.5 + 0.375 + 0.375$

$= 1.75$ bits.

Given codewords: 0, 10, 110, 111.

$L = \Sigma P(x) \times \text{Length}(x)$

$= (1/2 \times 1) + (1/4 \times 2) + (1/8 \times 3) + (1/8 \times 3)$

$= 1.75$ bits.

TASK 6: KL Divergence:

Given, $P = (0.5, 0.25, 0.25)$

$Q = (1/3, 1/3, 1/3)$

Formulae: $KL(P||Q) = \Sigma P(x) \log_2 [P(x)/Q(x)]$

Substituting:

$= 0.5 \log_2(1.5) + 0.25 \log_2(0.75) + 0.25 \log_2(0.75)$

$= 0.0849$ bits.

$KL(Q||P) = \Sigma Q(x) \log_2 [Q(x)/P(x)]$

$= (1/3)\log_2(2/3) + (1/3)\log_2(4/3) + (1/3)\log_2(4/3)$

$= 0.08170$ bits.

$KL(P||Q) \neq KL(Q||P)$.

7) Task 7: Entropy & Mutual Information:

Given,

$P(0,0) = 0.4$

$P(0,1) = 0.1$

$P(1,0) = 0.1$

$P(1,1) = 0.4$

$P(A=0) = 0.5$

$P(A=1) = 0.5$

$H(B) = -\sum P(B) \log_2 P(B)$

$= -[0.5 \log_2 0.5 + 0.5 \log_2 0.5]$

$= 1$ bit.

Conditional Entropy:

$H(A|B) = H(A,B) - H(B)$

$= 1.7219 - 1$

$= 0.7219$ bits.

Mutual Information:

$I(A;B) = H(B) - H(A|B)$

$= 1 - 0.7219$

$= 0.2781$ bits.

8) TASK 8: Bayesian Decision Theory:

Given:

$\mu_1 = 2, \mu_2 = 6$

$\sigma^2 = 4$

$P(\omega_1) = 0.6$

$P(\omega_2) = 0.4$

$x^* = (\mu_1 + \mu_2)/2 + [\sigma^2/(\mu_2 - \mu_1)] \ln[P(\omega_1)/P(\omega_2)]$

$= (2+6)/2 + (4/4) \ln(0.6/0.4)$

$= 4 + \ln(1.5) = 4.4055$

$P(\text{error}) = P(\omega_1)P(x > x^*|\omega_1) + P(\omega_2)P(x < x^*|\omega_2)$

$z_1 = (4.4055 - 2)/2 = 1.2027$

$z_2 = (4.4055 - 6)/2 = -0.7973$

$\Rightarrow 15.38\%$.

Risk and MAP decision

Given:

$P(\omega_1|x) = 0.3$

$P(\omega_2|x) = 0.7$

loss values:

$\lambda(\alpha_1|\omega_1) = 8$ $\quad \lambda(\alpha_1|\omega_2) = 50$

$\lambda(\alpha_2|\omega_1) = 0$ $\quad \lambda(\alpha_2|\omega_2) = 0$

a) Calculate risk of α_1

$R(\alpha_1|x) = 8(0.3) + 0(0.7)$

$= 2.4$

b) Calculate Risk of α_2

$R(\alpha_2|x) = 50(0.3) + 0.(0.7)$

$= 15$

c) Min Risk Decision:

$P(\omega_1|x) = 0.3$, $P(\omega_2|x) = 0.7$

10) TASK 10: Polynomial Regression & Overfitting

Given

Degree	Training MSE
1	9.0
2	4.1
3	2.2
5	0.9
9	0.1

a) Deg 1 : 9.0

Deg 2 : 4.1

Deg 3 : 2.2

Deg 4 : 0.9

Deg 5 : 0.1

Conclusion: Degree 9 has the lowest training MSE, but it is not necessarily the best model.

11) TASK 11: Bayesian Estimation of coin bias.

Given:
Number of tosses (n) = 10
Number of heads (h) = 7
Number of tails = 3
Prior distribution =
θ ~ Beta (2,2)

θ(ML) = No. of heads/Total tosses
= 7/10
= 0.7

2. ML Estimate
θ(ML) = 30/100
= 0.30

2. Map Estimate
Beta (72,32)
= Beta (72,32)
= 71/102
= 0.696

12) TASK 12: Early-Warning System for Sepsis prediction.

Given:
1000 patients, 5% sepsis, sentivity = 0.90, specificity = 0.80

Sepsis = 950
TP = 50 × 0.90 = 45
FN = 50 - 45 = 5
TN = 950 × 0.80 = 760
FP = 950 - 760 = 190
PPV = TP / (TP+FP) = 19.15%
Policy = P = 5/(100+5)
= 0.0476
= 4.76%